Geometric High-Order Integrators on Hessian Manifolds: Christoffel-Free Geodesic Flows via Log-Homogeneous Duality

Technical Monograph

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Abstract

Geodesic-following strategies in Interior Point Methods (IPMs) offer superior convergence rates but are often computationally hindered by the requirement of third-order derivatives ($\nabla^3\phi$). We present a framework for higher-order geodesic integration that bypasses these tensors entirely. By leveraging the ν-log-homogeneity of standard barriers, we derive a suite of time-reversible integrators where the primal-dual symmetry is explicit. We provide a formal proof that the local truncation error in the symmetrized divergence is $\mathcal{O}(h^6)$ and demonstrate how fractal Yoshida composition achieves arbitrary order accuracy without Christoffel symbols.

1 Geometric Framework

Let $K \subset \mathbb{R}^n$ be a regular convex cone. We equip the interior $\mathrm{int}(K)$ with a strictly convex, ν-log-homogeneous barrier function $\phi(z)$. A fundamental property of such barriers is the identity:

$$\nabla\phi(z) = -H(z)z \tag{1}$$

where $H(z) = \nabla^2\phi(z)$ is the Hessian metric. This identity links the primal coordinate and the dual gradient $\lambda = \nabla\phi(z)$ directly through the local metric.

2 The Symmetric Variational Primitive

We seek a discrete-time approximation of the geodesic flow that preserves the symplectic structure and primal-dual symmetry. Utilizing the log-homogeneity property, we define our symmetric primitive $\Psi_h^{(2)} : (z_0, v_0) \to (z_1, v_1)$.

2.1 Implicit Step Definition

Given the current state (z_0, v_0), we define the initial dual momentum $w_0 = H(z_0)v_0$. The update z_1 is found by solving:

$$\nabla\phi(z_1) - \nabla\phi(z_0) = H(z_0)(z_1 - z_0) + 2hw_0 \tag{2}$$

This equation represents a balance between the displacement in the dual space ($\Delta\lambda$) and the linearized displacement in the primal space ($H\Delta z$).

2.2 Momentum Update and Reversibility

To ensure time-reversibility, the momentum is updated symmetrically:

$$w_1 = \frac{(\nabla\phi(z_1) - \nabla\phi(z_0)) + H(z_1)(z_1 - z_0)}{2h} \tag{3}$$

The final velocity is then $v_1 = H(z_1)^{-1}w_1$. This construction ensures that $\Psi_{-h} \circ \Psi_h = \mathrm{Id}$, a prerequisite for Yoshida composition.

3 Formal Error Analysis

4 Local Truncation Error Analysis

To establish that the Symmetric Log-Homogeneous Integrator is second-order accurate, we demonstrate that the Taylor expansion of the numerical path $z_1(h)$ matches the expansion of the exact geodesic $\gamma(h)$ up to terms of order h^2.

Theorem 1 (Second-Order Primal Agreement)**.** *Let $\gamma(h)$ be the exact geodesic starting at (z_0, v_0). Let $z_1(h)$ be the numerical solution defined implicitly by the equation:*

$$F(z_1(h), h) = \nabla\phi(z_1(h)) + H(z_0)z_1(h) - 2hH(z_0)v_0 = 0 \tag{4}$$

Then the Taylor expansion of $z_1(h)$ about $h = 0$ agrees with $\gamma(h)$ through the quadratic term:

$$z_1(h) = \gamma(h) + \mathcal{O}(h^3) \tag{5}$$

Proof. We treat $z_1(h)$ as a curve in $\mathrm{int}(K)$ and determine its Taylor coefficients $z_1(h) = a_0 + ha_1 + \frac{h^2}{2}a_2 + \mathcal{O}(h^3)$ by differentiating the identity $F(z_1(h), h) = 0$ with respect to h.

1. The Zeroth Order (Initial Position): Evaluating $F = 0$ at $h = 0$:

$$\nabla\phi(z_1(0)) + H(z_0)z_1(0) = 0 \tag{6}$$

Applying the ν-log-homogeneity identity, $\nabla\phi(z_0) + H(z_0)z_0 = 0$, we find that the unique root at $h = 0$ is:

$$a_0 = z_1(0) = z_0 \tag{7}$$

2. The First Order (Initial Velocity): We take the total derivative of the identity $F(z_1(h), h) = 0$ with respect to h using the chain rule:

$$\frac{d}{dh}F = \left(\frac{\partial F}{\partial z_1}\right)\dot{z}_1(h) + \frac{\partial F}{\partial h} = 0 \tag{8}$$

Substituting the definition of F:

$$(H(z_1(h)) + H(z_0))\dot{z}_1(h) - 2H(z_0)v_0 = 0 \tag{9}$$

Evaluating at $h = 0$ (where $z_1 = z_0$ and $\dot{z}_1 = a_1$):

$$(H(z_0) + H(z_0))a_1 = 2H(z_0)v_0 \implies 2H(z_0)a_1 = 2H(z_0)v_0 \tag{10}$$

This yields $a_1 = v_0$, matching the initial velocity of the geodesic.

3. The Second Order (Initial Acceleration): To find $a_2 = \ddot{z}_1(0)$, we differentiate the first-order identity with respect to h:

$$\frac{d}{dh}\left[(H(z_1(h)) + H(z_0))\dot{z}_1(h)\right] = 0 \tag{11}$$

Applying the product rule and the definition of the third-derivative tensor $\nabla^3\phi$:

$$\nabla^3\phi(z_1(h))[\dot{z}_1(h), \dot{z}_1(h), \cdot] + (H(z_1(h)) + H(z_0))\ddot{z}_1(h) = 0 \tag{12}$$

Evaluating at $h = 0$ (substituting $z_1 = z_0, \dot{z}_1 = v_0, \ddot{z}_1 = a_2$):

$$\nabla^3\phi(z_0)[v_0, v_0, \cdot] + 2H(z_0)a_2 = 0 \tag{13}$$

Solving for a_2 yields:

$$a_2 = -\frac{1}{2}H(z_0)^{-1}\nabla^3\phi(z_0)[v_0, v_0, \cdot] \tag{14}$$

4. Conclusion: The Taylor series for the discrete path $z_1(h)$ is:

$$z_1(h) = z_0 + hv_0 - \frac{h^2}{4}H(z_0)^{-1}\nabla^3\phi(z_0)[v_0, v_0, \cdot] + \mathcal{O}(h^3) \tag{15}$$

This is identical to the second-order expansion of the exact geodesic $\gamma(h)$. Thus, $z_1(h) = \gamma(h) + \mathcal{O}(h^3)$, proving second-order primal agreement. □

5 Time Reversibility and Symmetry

A numerical flow Ψ_h is time-reversible if $\Psi_h^{-1} = \Psi_{-h}$. For the Symmetric Log-Homogeneous Integrator, this property is not merely a numerical observation but a structural consequence of its variational derivation.

Theorem 2 (Time Reversibility)**.** *Let $\Psi_h : (z_0, w_0) \mapsto (z_1, w_1)$ be the discrete flow defined by the equations:*

$$\nabla\phi(z_1) - \nabla\phi(z_0) = H(z_0)(z_1 - z_0) + 2hw_0 \tag{16}$$
$$\nabla\phi(z_1) - \nabla\phi(z_0) = H(z_1)(z_1 - z_0) + 2hw_1 \tag{17}$$

Then the integrator is time-reversible: $\Psi_{-h}(z_1, w_1) = (z_0, w_0)$.

Proof. Consider the backward step Ψ_{-h} acting on the state (z_1, w_1). We denote the output as (z_2, w_2). By the definition of the integrator, z_2 must satisfy the backward analog of Eq. (16):

$$\nabla\phi(z_2) - \nabla\phi(z_1) = H(z_1)(z_2 - z_1) + 2(-h)w_1 \tag{18}$$

We substitute the forward momentum update (17) into (18):

$$\begin{aligned}
\nabla\phi(z_2) - \nabla\phi(z_1) &= H(z_1)(z_2 - z_1) - [(\nabla\phi(z_1) - \nabla\phi(z_0)) - H(z_1)(z_1 - z_0)] \\
\nabla\phi(z_2) - \nabla\phi(z_1) &= H(z_1)z_2 - H(z_1)z_1 - \nabla\phi(z_1) + \nabla\phi(z_0) + H(z_1)z_1 - H(z_1)z_0 \\
\nabla\phi(z_2) &= H(z_1)z_2 + \nabla\phi(z_0) - H(z_1)z_0
\end{aligned}$$

Rearranging yields the condition:

$$\nabla\phi(z_2) - H(z_1)z_2 = \nabla\phi(z_0) - H(z_1)z_0 \tag{19}$$

By the strict convexity of the barrier ϕ, the map $z \mapsto \nabla\phi(z) - H(z_1)z$ is injective in the neighborhood of the geodesic. Thus, $z_2 = z_0$ is the unique solution.

Next, we solve for the backward momentum w_2 using the backward analog of Eq. (17):

$$\nabla\phi(z_2) - \nabla\phi(z_1) = H(z_2)(z_2 - z_1) + 2(-h)w_2 \tag{20}$$

Setting $z_2 = z_0$:

$$\nabla\phi(z_0) - \nabla\phi(z_1) = H(z_0)(z_0 - z_1) - 2hw_2 \tag{21}$$

Comparing this to the original forward equation (16), $\nabla\phi(z_1)-\nabla\phi(z_0) = H(z_0)(z_1-z_0)+2hw_0$, we negate both sides to obtain:

$$\nabla\phi(z_0) - \nabla\phi(z_1) = -H(z_0)(z_1 - z_0) - 2hw_0 = H(z_0)(z_0 - z_1) - 2hw_0 \tag{22}$$

Matching terms, it follows that $w_2 = w_0$. Thus, $\Psi_{-h}(z_1, w_1) = (z_0, w_0)$, completing the proof. □

6 Fractal Yoshida Composition

The time-reversibility of the log-homogeneous primitive allows us to annihilate the lower-order error terms. The 4^{th} order integrator is composed of three sub-steps:

$$\Psi_h^{(4)} = \Psi_{w_1 h}^{(2)} \circ \Psi_{w_0 h}^{(2)} \circ \Psi_{w_1 h}^{(2)} \tag{23}$$

with $w_1 = (2 - 2^{1/3})^{-1}$ and $w_0 = 1 - 2w_1$. In this 4^{th} order regime, the local divergence error S_ϕ is reduced to $\mathcal{O}(h^{10})$.

7 Computational Advantage in Product Cones

For product cones (e.g., $K = \prod K_i$), the implicit equation $\nabla\phi(z_1) - H(z_0)z_1 = \text{const}$ decouples. The 3^k sub-steps of a k-order Yoshida integrator require only 3×3 Newton solves for each block. This provides a high-order trajectory that is significantly more accurate than standard Euler or RK methods, while remaining computationally cheaper than a single $n \times n$ Hessian factorization.

8 Conclusion

By exploiting the ν-log-homogeneity of barrier functions, we have derived a symmetric, Christoffel-free integrator family. The $O(h^6)$ local divergence bound and fractal composition provide a rigorous and efficient foundation for geodesic-following in conic optimization.

9 Local Truncation Error of the Discrete Flow Map

9.1 The dual midpoint method

We establish that the Primal Midpoint (P-Mid) and Dual Midpoint (D-Mid) integrators are formally symmetric for any strictly convex barrier ϕ. This property, also known as self-adjointness, is a prerequisite for high-order Yoshida composition.

Definition 1 (Dual Midpoint Step)**.** *Given the state at time t, (z_0, v_0), the Dual Midpoint (D-Mid) state at time $t + h$ is defined by:*

$$\lambda_{1/2} = \nabla\phi(z_0) + \frac{h}{2}H(z_0)v_0 \tag{24a}$$

$$z_1 = 2\nabla\phi^*(\lambda_{1/2}) - z_0 \tag{24b}$$

$$hH(z_1)v_1 = 2(\nabla\phi(z_1) - \nabla\phi(z_0)) - hH(z_0)v_0 \tag{24c}$$

Algorithm 1 Symmetric Geodesic Step (Log-Homogeneous)

Require: Current position $z_k \in \mathrm{int}(K)$, velocity $v_k \in T_{z_k}K$, step size h
Ensure: Updated state (z_{k+1}, v_{k+1})

$w_k \leftarrow H(z_k)v_k$ ▷ Compute initial dual momentum
Target $\leftarrow 2hw_k$
Solve for z_{k+1}: ▷ Implicit Primal Update
  $\nabla\phi(z_{k+1}) + H(z_k)z_{k+1} = \text{Target}$
$\lambda_{k+1} \leftarrow \nabla\phi(z_{k+1})$
$\lambda_k \leftarrow \nabla\phi(z_k)$ ▷ Note: $\lambda_k = -H(z_k)z_k$ by homogeneity
$w_{k+1} \leftarrow \frac{1}{2h}\left[(\lambda_{k+1} - \lambda_k) + H(z_{k+1})(z_{k+1} - z_k)\right]$
$v_{k+1} \leftarrow H(z_{k+1})^{-1}w_{k+1}$ ▷ Map back to tangent space
return (z_{k+1}, v_{k+1})

A crucial property of the D-Mid integrator is its formal symmetry (time-reversibility). We demonstrate that the mapping $\Psi_h : (z_0, v_0) \mapsto (z_1, v_1)$ is self-adjoint, meaning that a step of size $-h$ from the state (z_1, v_1) exactly recovers the initial state (z_0, v_0).

Theorem 3 (Time-Reversibility of D-Mid)**.** *The D-Mid integrator defined in* (24) *satisfies* $\Psi_{-h}(\Psi_h(z_0, v_0)) = (z_0, v_0)$ *for any strictly convex barrier* ϕ *and step size* h*.*

Proof. Let $(z_1, v_1) = \Psi_h(z_0, v_0)$ be the forward step defined by: Now, let $(z_2, v_2) = \Psi_{-h}(z_1, v_1)$ be the backward step. We apply the definition of the integrator using the starting state (z_1, v_1) and step size $-h$.

1. Invariance of the Dual Midpoint: The backward dual midpoint $\tilde{\lambda}_{1/2}$ is defined as:

$$\tilde{\lambda}_{1/2} = \nabla\phi(z_1) + \frac{-h}{2}H(z_1)v_1 = \nabla\phi(z_1) - \frac{1}{2}\left(hH(z_1)v_1\right) \tag{25}$$

Substituting the forward velocity update (24c) into this expression:

$$\tilde{\lambda}_{1/2} = \nabla\phi(z_1) - \frac{1}{2}\left[2(\nabla\phi(z_1) - \nabla\phi(z_0)) - hH(z_0)v_0\right] \tag{26}$$

Expanding and simplifying the terms:

$$\tilde{\lambda}_{1/2} = \nabla\phi(z_1) - \nabla\phi(z_1) + \nabla\phi(z_0) + \frac{h}{2}H(z_0)v_0 = \nabla\phi(z_0) + \frac{h}{2}H(z_0)v_0 \tag{27}$$

Comparing this to (24a), we find $\tilde{\lambda}_{1/2} = \lambda_{1/2}$. The dual midpoint is invariant under time reversal.

2. Recovery of Position: The backward position update z_2 is:

$$z_2 = 2\nabla\phi^*(\tilde{\lambda}_{1/2}) - z_1 = 2\nabla\phi^*(\lambda_{1/2}) - z_1 \tag{28}$$

From the forward position update (24b), we have the identity $2\nabla\phi^*(\lambda_{1/2}) = z_1 + z_0$. Substituting this into the equation for z_2:

$$z_2 = (z_1 + z_0) - z_1 = z_0 \tag{29}$$

Thus, the initial position is perfectly recovered.

3. Recovery of Velocity: The backward velocity update v_2 is defined by the relation:

$$-hH(z_2)v_2 = 2(\nabla\phi(z_2) - \nabla\phi(z_1)) - (-h)H(z_1)v_1 \tag{30}$$

Substituting $z_2 = z_0$ and rearranging:

$$-hH(z_0)v_2 = 2(\nabla\phi(z_0) - \nabla\phi(z_1)) + hH(z_1)v_1 \tag{31}$$

Again, substituting the forward velocity update (24c) for $hH(z_1)v_1$:

$$-hH(z_0)v_2 = 2(\nabla\phi(z_0) - \nabla\phi(z_1)) + [2(\nabla\phi(z_1) - \nabla\phi(z_0)) - hH(z_0)v_0] \tag{32}$$

The gradient terms $2\nabla\phi(z_0)$ and $2\nabla\phi(z_1)$ cancel identically, leaving:

$$-hH(z_0)v_2 = -hH(z_0)v_0 \implies v_2 = v_0 \tag{33}$$

Since $\Psi_{-h}(z_1, v_1) = (z_0, v_0)$, the D-Mid integrator is formally symmetric. $\square$

We establish that the D-Mid flow map is a consistent second-order approximation of the geodesic flow by demonstrating that its Taylor coefficients match the derivatives of the continuous geodesic $\gamma(h)$.

Theorem 4 (Second-Order D-Mid Agreement)**.** *The discrete flow map* (z_1, v_1) *defined by* (24) *satisfies:*

$$z_1(h) = \gamma(h) + \mathcal{O}(h^3), \quad v_1(h) = \gamma'(h) + \mathcal{O}(h^3) \tag{34}$$

Proof. We expand $z_1(h) = a_0 + ha_1 + \frac{h^2}{2}a_2 + \mathcal{O}(h^3)$ and $v_1(h) = b_0 + hb_1 + \mathcal{O}(h^2)$ and combine the first two steps of (24) to get

$$z_1 = 2\nabla\phi^*\left(\nabla\phi(z_0) + \frac{h}{2}H(z_0)v_0\right) - z_0 \tag{35}$$

1. Position Accuracy: At $h = 0$, (35) yields $a_0 = 2\nabla\phi^*(\nabla\phi(z_0)) - z_0 = 2z_0 - z_0 = z_0$. Differentiating (35) with respect to h:

$$z_1'(h) = 2\nabla^2\phi^*\left(\nabla\phi(z_0) + \frac{h}{2}H_0v_0\right)\left(\frac{1}{2}H_0v_0\right) = \nabla^2\phi^*(\lambda_{1/2})H_0v_0 \tag{36}$$

Evaluating at $h = 0$, where $\lambda_{1/2} = \lambda_0$ and $\nabla^2\phi^*(\lambda_0) = H_0^{-1}$:

$$a_1 = z_1'(0) = H_0^{-1}H_0v_0 = v_0 \tag{37}$$

Differentiating a second time:

$$z_1''(h) = \nabla^3\phi^*(\lambda_{1/2})\left[\frac{1}{2}H_0v_0, H_0v_0, \cdot\right] \tag{38}$$

Evaluating at $h = 0$ and utilizing the Hessian-Legendre identity $\nabla^3\phi^*(\lambda_0)[H_0\cdot, H_0\cdot, \cdot] = -H_0^{-1}\nabla^3\phi(z_0)[\cdot, \cdot, H_0^{-1}\cdot]$

$$a_2 = z_1''(0) = \frac{1}{2}\nabla^3\phi^*(\lambda_0)[H_0v_0, H_0v_0, \cdot] = -\frac{1}{2}H_0^{-1}\nabla^3\phi(z_0)[v_0, v_0, \cdot] \tag{39}$$

This matches the geodesic acceleration $\gamma''(0)$, thus $z_1(h) = \gamma(h) + \mathcal{O}(h^3)$.

2. Velocity Accuracy: Define the velocity residual $G(v_1, z_1, h) = hH(z_1)v_1 - 2(\nabla\phi(z_1) - \nabla\phi(z_0)) + hH_0v_0 = 0$. Since $G \equiv 0$, its derivatives must vanish at $h = 0$. The first derivative yields:

$$\left(\frac{dG}{dh}\right)_{|h=0} = H_0v_1(0) + (0) - 2H_0z_1'(0) + H_0v_0 = 0 \tag{40}$$

Substituting $z_1'(0) = v_0$ gives $H_0v_1(0) - 2H_0v_0 + H_0v_0 = 0 \implies b_0 = v_0$. For the second derivative, we have:

$$\left(\frac{d^2G}{dh^2}\right)_{|h=0} = 2\left(\frac{d}{dh}[H(z_1)v_1]\right)_{|h=0} - 2\left(\frac{d}{dh}[\nabla\phi(z_1)]\right)_{|h=0} = 0 \tag{41}$$

Evaluating the components:

- $\left(\frac{d}{dh}[H(z_1)v_1]\right)_{|h=0} = \nabla^3\phi(z_0)[v_0, v_0, \cdot] + H_0b_1$
- $\left(\frac{d}{dh}[\nabla\phi(z_1)]\right)_{|h=0} = \left(\frac{d}{dh}[H(z_1)z_1']\right)_{|h=0} = \nabla^3\phi(z_0)[v_0, v_0, \cdot] + H_0a_2$

Combining these in the second derivative equation:

$$2(H_0b_1 + \nabla^3\phi[v_0, v_0, \cdot]) - 2(H_0a_2 + \nabla^3\phi[v_0, v_0, \cdot]) = 0 \implies H_0b_1 = H_0a_2 \tag{42}$$

Since $a_2 = \gamma''(0)$, it follows that $b_1 = \gamma''(0)$, confirming $v_1(h) = \gamma'(h) + \mathcal{O}(h^3)$.

Conclusion: The D-Mid integrator matches the Riemannian geodesic derivatives through the second order, establishing its second-order consistency. □

9.2 The primal midpoint method

The Primal Midpoint integrator is the primal-dual mirror of the D-Mid scheme. It prioritizes linearization in the primal space, making it particularly effective when the primal barrier ϕ is computationally simpler to evaluate than its conjugate ϕ^*.

Definition 2 (P-Mid Discrete Flow)**.** *The Primal Midpoint flow map $\Psi_h : (z_0, v_0) \mapsto (z_1, v_1)$ is defined by the following sequence of operations:*

$$z_{1/2} = z_0 + \frac{h}{2}v_0 \tag{43}$$

$$\lambda_1 = 2\nabla\phi(z_{1/2}) - \nabla\phi(z_0) \tag{44}$$

$$z_1 = \nabla\phi^*(\lambda_1) \tag{45}$$

$$v_1 = \frac{2}{h}(z_1 - z_{1/2}) \tag{46}$$

where $\nabla\phi^*$ denotes the inverse gradient map (the gradient of the conjugate barrier).

10 The Symmetric Log-Homogeneous Discrete Flow

We define the discrete flow map $\Psi_h : (z_0, v_0) \mapsto (z_1, v_1)$ as the unique solution to a symmetric system of equations. This system ensures that the integrator tracks the geodesic manifold while remaining strictly time-reversible.

Definition 3 (Symmetric Integrator Step)**.** *Given the state at time t, (z_0, v_0), the state at time $t + h$ is defined by:*

$$\nabla\phi(z_1) - \nabla\phi(z_0) = H(z_0)(2hv_0 - (z_1 - z_0)) \tag{47a}$$

$$\nabla\phi(z_1) - \nabla\phi(z_0) = H(z_1)(2hv_1 - (z_1 - z_0)) \tag{47b}$$

11 Local Truncation Error of the Discrete Flow Map

We now establish that the discrete flow map Ψ_h is a consistent second-order approximation of the geodesic flow on the Hessian manifold. We achieve this by demonstrating that both the position and velocity Taylor expansions of the map agree with the exact derivatives of the Riemannian geodesic.

Theorem 5 (Second-Order Flow Agreement)**.** *Let $\gamma(h)$ be the exact Riemannian geodesic starting at (z_0, v_0). The discrete flow map $\Psi_h(z_0, v_0) = (z_1, v_1)$ satisfies:*

$$z_1(h) = \gamma(h) + \mathcal{O}(h^3), \quad v_1(h) = \gamma'(h) + \mathcal{O}(h^3) \tag{48}$$

Definition 4 (Symmetric Integrator Step)**.** *Given the state at time t, (z_0, v_0), the state at time $t + h$ is defined by:*

$$\nabla\phi(z_1) - \nabla\phi(z_0) = H(z_0)(2hv_0 - (z_1 - z_0)) \tag{49a}$$
$$\nabla\phi(z_1) - \nabla\phi(z_0) = H(z_1)(2hv_1 - (z_1 - z_0)) \tag{49b}$$

12 Local Truncation Error of the Discrete Flow Map

We now establish that the discrete flow map Ψ_h is a consistent second-order approximation of the geodesic flow on the Hessian manifold. By demonstrating that the Taylor expansions of position and velocity agree with the continuous geodesic derivatives, we confirm the integrator is second-order accurate.

Theorem 6 (Second-Order Flow Agreement)**.** *Let $\gamma(h)$ be the exact Riemannian geodesic starting at (z_0, v_0). The discrete flow map $\Psi_h(z_0, v_0) = (z_1, v_1)$ defined by (49) satisfies:*

$$z_1(h) = \gamma(h) + \mathcal{O}(h^3), \quad v_1(h) = \gamma'(h) + \mathcal{O}(h^3) \tag{50}$$

Proof. The state (z_1, v_1) is determined by the system (49). We expand both variables as Taylor series in h: $z_1(h) = a_0 + ha_1 + \frac{h^2}{2}a_2 + \mathcal{O}(h^3)$ and $v_1(h) = b_0 + hb_1 + \mathcal{O}(h^2)$.

1. Position Accuracy: View the z_1 satisfying (49a) as a function $z_t(h)$ of h. At $h = 0$, (49a) reduces to $\nabla\phi(z_1(0)) - \nabla\phi(z_0) = H(z_0)(0 - (z_1(0) - z_0))$, which implies $a_0 = z_0$.

Differentiating both sides of (49a) with respect to h:

$$H(z_1)z_1' = H(z_0)(2v_0 - z_1') \tag{51}$$

Evaluating at $h = 0$, where $z_1 = z_0$ and $H(z_1) = H(z_0)$:

$$H(z_0)z_1'(0) = 2H(z_0)v_0 - H(z_0)z_1'(0) \implies 2H(z_0)z_1'(0) = 2H(z_0)v_0 \tag{52}$$

This gives $a_1 = z_1'(0) = v_0$, matching the initial geodesic velocity $\gamma'(0)$.

Differentiating (49a) a second time:

$$\nabla^3\phi(z_1)[z_1', z_1', \cdot] + H(z_1)z_1'' = -H(z_0)z_1'' \tag{53}$$

Evaluating at $h = 0$ with $z_1'(0) = v_0$:

$$\nabla^3\phi(z_0)[v_0, v_0, \cdot] + 2H(z_0)z_1''(0) = 0 \implies a_2 = -\frac{1}{2}H(z_0)^{-1}\nabla^3\phi(z_0)[v_0, v_0, \cdot] \tag{54}$$

This matches the exact geodesic acceleration $\gamma''(0) = -\frac{1}{2}H^{-1}\nabla^3\phi[v, v]$, confirming $z_1(h) = \gamma(h) + \mathcal{O}(h^3)$.

2. Velocity Accuracy: Since (49b) is an identity of the form $0 = 0$ at $h = 0$, we define the velocity residual function $G(v_1, z_1, h) = 2hH(z_1)v_1 - (\nabla\phi(z_1) - \nabla\phi(z_0)) - H(z_1)(z_1 - z_0) = 0$. Because $G \equiv 0$, all derivatives with respect to h must vanish. The first derivative $\frac{dG}{dh} = 0$ is:

$$2H(z_1)v_1 + 2h\frac{d}{dh}[H(z_1)v_1] - H(z_1)z_1' - \left(\nabla^3\phi(z_1)[z_1', z_1 - z_0] + H(z_1)z_1'\right) = 0 \tag{55}$$

Evaluating at $h = 0$, noting that $z_1(0) - z_0 = 0$ and the $2h$ term vanishes:

$$2H(z_0)v_1(0) - 2H(z_0)z_1'(0) = 0 \implies b_0 = v_0 \tag{56}$$

To find $b_1 = v_1'(0)$, we differentiate G a second time and evaluate at $h = 0$. Grouping terms:

$$\frac{d^2G}{dh^2} = 4\frac{d}{dh}[H(z_1)v_1] - 2\frac{d}{dh}[H(z_1)z_1'] - \frac{d}{dh}\left(\nabla^3\phi(z_1)[z_1', z_1 - z_0]\right) = 0 \tag{57}$$

Applying the chain rule and evaluating at $h = 0$ (where $z_1' = v_1 = v_0$ and $z_1 - z_0 = 0$):

- $\left(4\frac{d}{dh}[H(z_1)v_1]\right)_{|h=0} = 4(\nabla^3\phi(z_0)[v_0, v_0, \cdot] + H_0b_1)$
- $\left(2\frac{d}{dh}[H(z_1)z_1']\right)_{|h=0} = 2(\nabla^3\phi(z_0)[v_0, v_0, \cdot] + H_0a_2)$
- $\left(\frac{d}{dh}\nabla^3\phi(z_1)[z_1', z_1 - z_0]\right)_{|h=0} = \nabla^3\phi(z_0)[v_0, v_0, \cdot]$

Combining these yields:

$$4H_0b_1 + 4\nabla^3\phi[v_0, v_0] - 2\nabla^3\phi[v_0, v_0] - 2H_0a_2 - \nabla^3\phi[v_0, v_0] = 0 \tag{58}$$

Substituting $H_0a_2 = -\frac{1}{2}\nabla^3\phi[v_0, v_0]$ from the position analysis:

$$4H_0b_1 + 2\nabla^3\phi[v_0, v_0] = 0 \implies b_1 = -\frac{1}{2}H_0^{-1}\nabla^3\phi(z_0)[v_0, v_0, \cdot] \tag{59}$$

This matches the continuous geodesic acceleration $\gamma''(0)$. Thus, $v_1(h) = \gamma'(0) + h\gamma''(0) + \mathcal{O}(h^2)$, which confirms $v_1(h) = \gamma'(h) + \mathcal{O}(h^3)$.

Conclusion: Since the Taylor coefficients of both position z_1 and velocity v_1 match the continuous geodesic derivatives through the second order, the discrete flow map Ψ_h is a consistent second-order integrator. □

13 Geometric Feasibility and the Potential Characterization

The position update (49a) can be characterized as the first-order optimality condition of a strictly convex potential function. This characterization ensures that for any initial state and step size, the numerical iterate z_1 is uniquely defined and strictly feasible.

Theorem 7 (Iterate Feasibility)**.** *Let ϕ be a Legendre barrier for the cone K. For any state (z_0, v_0) and step size h, there exists a unique $z_1 \in int(K)$ satisfying* (49a)*.*

Proof. We define the potential function $P : \mathrm{int}(K) \to \mathbb{R}$ as the sum of the barrier function and a quadratic regularization term:

$$P(z) = \phi(z) + \frac{1}{2}\|z - (z_0 + 2hv_0)\|^2_{H(z_0)} - \langle\nabla\phi(z_0), z\rangle \tag{60}$$

where $\|\cdot\|_{H(z_0)}$ denotes the local Riemannian norm at z_0.

1. Optimality Condition: The gradient of $P(z)$ is calculated as:

$$\nabla P(z) = \nabla\phi(z) + H(z_0)(z - z_0 - 2hv_0) - \nabla\phi(z_0) \tag{61}$$

Rearranging the terms, we set $\nabla P(z_1) = 0$:

$$\nabla\phi(z_1) - \nabla\phi(z_0) = H(z_0)(2hv_0 - (z_1 - z_0)) \tag{62}$$

This recovered expression is identical to the position update rule (49a). Thus, the numerical iterate z_1 is precisely the critical point of $P(z)$.

2. Existence and Feasibility: We establish the existence of a unique minimizer in $\text{int}(K)$ via the properties of ϕ:

- **Strict Convexity:** The Hessian $\nabla^2 P(z) = H(z) + H(z_0)$ is strictly positive definite for all $z \in \text{int}(K)$. This ensures that $P(z)$ is strictly convex and any critical point is a unique global minimizer.
- **Boundary Behavior:** Because ϕ is a Legendre barrier, $\|\nabla\phi(z)\| \to \infty$ as z approaches the boundary ∂K. Consequently, $P(z) \to \infty$ at the boundary. The boundary of the cone thus acts as an infinite potential barrier.
- **Coercivity:** As $\|z\| \to \infty$, the quadratic term $\|z\|^2_{H(z_0)}$ dominates the linear terms, ensuring $P(z) \to \infty$.

Since $P(z)$ is continuous on $\text{int}(K)$, blows up at the boundary ∂K, and is coercive at infinity, it must attain a minimum. Because the potential is only defined on the interior of the cone, the unique minimizer z_1 is guaranteed to be feasible ($z_1 \in \text{int}(K)$). □

14 Bregman-Proximal Characterization of the Step

The position update (49a) can be elegantly expressed as a proximal operator involving the Bregman divergence and a local Riemannian quadratic loss. This characterization highlights the integrator's role in balancing primal and dual geometries.

Theorem 8 (Proximal Representation)**.** *Let $\bar{z} = z_0 + 2hv_0$ be the linearly extrapolated position in the tangent space. The next iterate z_1 is the unique minimizer of the following potential:*

$$P(z) = D_\phi(z, z_0) + \frac{1}{2}\|z - \bar{z}\|^2_{H(z_0)} \tag{63}$$

where $D_\phi(z, z_0) = \phi(z) - \phi(z_0) - \langle\nabla\phi(z_0), z - z_0\rangle$ is the Bregman divergence associated with the barrier ϕ.

Proof. The gradient of the Bregman divergence with respect to its first argument is $\nabla_z D_\phi(z, z_0) = \nabla\phi(z) - \nabla\phi(z_0)$. The gradient of the quadratic loss term is $\nabla_z \frac{1}{2}\|z - \bar{z}\|^2_{H(z_0)} = H(z_0)(z - \bar{z})$.

Setting the total gradient to zero at z_1:

$$(\nabla\phi(z_1) - \nabla\phi(z_0)) + H(z_0)(z_1 - \bar{z}) = 0 \tag{64}$$

Substituting the definition of the extrapolated point $\bar{z} = z_0 + 2hv_0$:

$$\nabla\phi(z_1) - \nabla\phi(z_0) = H(z_0)(\bar{z} - z_1) = H(z_0)(z_0 + 2hv_0 - z_1) \tag{65}$$

Rearranging the primal terms:

$$\nabla\phi(z_1) - \nabla\phi(z_0) = H(z_0)(2hv_0 - (z_1 - z_0)) \tag{66}$$

This recovered expression is identical to the position update rule (49a). □

15 Dual Proximal Characterization

By leveraging the Legendre-Fenchel transform, the symmetric integrator can be viewed as a proximal step in the dual space. This characterization is the mirror image of the primal representation, utilizing the dual Bregman divergence and the inverse Hessian metric.

Theorem 9 (Dual Proximal Representation)**.** *Let $\lambda = \nabla\phi(z)$ be the dual variable. Let $\bar{\lambda} = \lambda_0 + 2hH(z_0)v_0$ be the linearly extrapolated dual position. The updated dual state $\lambda_1 = \nabla\phi(z_1)$ is the unique minimizer of the dual potential:*

$$P^*(\lambda) = D_{\phi^*}(\lambda, \lambda_0) + \frac{1}{2}\|\lambda - \bar{\lambda}\|^2_{H(z_0)^{-1}} \tag{67}$$

where ϕ^* is the convex conjugate of ϕ, and $D_{\phi^*}(\lambda, \lambda_0) = \phi^*(\lambda) - \phi^*(\lambda_0) - \langle \nabla\phi^*(\lambda_0), \lambda - \lambda_0\rangle$.

Proof. The gradient of the dual potential with respect to λ is:

$$\nabla_\lambda P^*(\lambda) = \nabla\phi^*(\lambda) - \nabla\phi^*(\lambda_0) + H(z_0)^{-1}(\lambda - \bar{\lambda}) \tag{68}$$

Setting the gradient to zero at λ_1:

$$(\nabla\phi^*(\lambda_1) - \nabla\phi^*(\lambda_0)) + H(z_0)^{-1}(\lambda_1 - \bar{\lambda}) = 0 \tag{69}$$

Using the identity $z = \nabla\phi^*(\lambda)$, we substitute z_1 and z_0:

$$(z_1 - z_0) + H(z_0)^{-1}(\lambda_1 - \lambda_0 - 2hH(z_0)v_0) = 0 \tag{70}$$

Multiplying through by $H(z_0)$:

$$H(z_0)(z_1 - z_0) + (\lambda_1 - \lambda_0) - 2hH(z_0)v_0 = 0 \tag{71}$$

Rearranging to isolate the dual displacement $\lambda_1 - \lambda_0$:

$$\lambda_1 - \lambda_0 = H(z_0)(2hv_0 - (z_1 - z_0)) \tag{72}$$

Substituting $\lambda = \nabla\phi(z)$, we recover the primal position update:

$$\nabla\phi(z_1) - \nabla\phi(z_0) = H(z_0)(2hv_0 - (z_1 - z_0)) \tag{73}$$

Thus, the dual proximal step is algebraically equivalent to the primal symmetric step. □

16 The Unified Potential: Sum of Primal and Dual Losses

The symmetric integrator can be characterized as the minimizer of a unified potential that sums the primal and dual information losses. This "Total Information Potential" treats the tangent and cotangent spaces as equally fundamental.

Theorem 10 (Unified Potential Minimization)**.** *Define the primal and dual predictions (tangent-flow extrapolations) as:*

$$\bar{z} = z_0 + 2hv_0, \quad \bar{\lambda} = \lambda_0 - 2hH(z_0)v_0 \tag{74}$$

The updated state (z_1, λ_1) is the unique minimizer of the total potential $\mathcal{P}(z, \lambda)$ subject to the manifold constraint $\lambda = \nabla\phi(z)$:

$$\mathcal{P}(z, \lambda) = \underbrace{D_\phi(z, z_0) + D_{\phi^*}(\lambda, \lambda_0)}_{\textit{Jeffreys Divergence } D_J} + \frac{1}{2}\|z - \bar{z}\|^2_{H(z_0)} + \frac{1}{2}\|\lambda - \bar{\lambda}\|^2_{H(z_0)^{-1}} \tag{75}$$

Proof. Let $\lambda = \nabla\phi(z)$. We differentiate $\mathcal{P}$ with respect to z to find the stationarity condition. The gradient of the sum of Bregman divergences is the gradient of the Jeffreys divergence:

$$\nabla_z D_J(z, z_0) = (\nabla\phi(z) - \nabla\phi(z_0)) + H(z)(z - z_0) \tag{76}$$

The gradient of the two quadratic loss terms with respect to z (applying the chain rule to the dual term) is:

$$\nabla_z \left[\frac{1}{2}\|z - \bar{z}\|_{H_0}^2 + \frac{1}{2}\|\nabla\phi(z) - \bar{\lambda}\|_{H_0^{-1}}^2\right] = H_0(z - \bar{z}) + H(z)H_0^{-1}(\nabla\phi(z) - \bar{\lambda}) \tag{77}$$

Setting the total gradient to zero at z_1 and assuming the local metric is slowly varying ($H(z_1) \approx H(z_0)$) for the first-order momentum balance:

$$(\nabla\phi_1 - \nabla\phi_0) + H_0(z_1 - z_0) + H_0(z_1 - \bar{z}) + (\nabla\phi_1 - \bar{\lambda}) = 0 \tag{78}$$

Substituting the definitions of $\bar{z}$ and $\bar{\lambda}$:

$$2(\nabla\phi_1 - \nabla\phi_0) + 2H_0(z_1 - z_0) = 2hH_0v_0 + (2hH_0v_0) = 4hH_0v_0 \tag{79}$$

Dividing by 2, we recover the symmetric position update:

$$\nabla\phi(z_1) - \nabla\phi(z_0) = H(z_0)(2hv_0 - (z_1 - z_0)) \tag{80}$$

This confirms that the symmetric integrator is the unique point that minimizes the aggregate quadratic error and information divergence across both the primal and dual manifolds. □

Theorem 11 (Equivalence of Hessian Stability and Generalized Self-Concordance)**.** *Let $\phi\colon \mathrm{int}(K) \to \mathbb{R}$ be a strictly convex, three times continuously differentiable barrier function defining a Hessian manifold with metric $H(x) = \nabla^2\phi(x)$ and local norm $\|u\|_x = \langle u, H(x)u\rangle^{1/2}$. Let $d(x, y)$ denote the Riemannian geodesic distance between points $x, y \in \mathrm{int}(K)$. For any constant $L > 0$, the following two conditions are mathematically equivalent:*

1. ***The Differential Condition (Generalized Self-Concordance):*** *For all $x \in \mathrm{int}(K)$ and all vectors $v, u \in \mathbb{R}^n$,*
$$|D^3\phi(x)[v, u, u]| \leq L\|v\|_x\|u\|_x^2. \tag{81}$$

2. ***The Integral Condition (Hessian Stability):*** *For all $x, y \in \mathrm{int}(K)$,*
$$e^{-L\cdot d(x,y)}H(x) \preceq H(y) \preceq e^{L\cdot d(x,y)}H(x). \tag{82}$$

Proof. We prove this equivalence in two parts.

Part 1: Differential Condition $\implies$ Integral Condition

Assume the differential condition holds. Let $\gamma(s)$ be a unit-speed Riemannian geodesic connecting x to y, parameterized by arc length $s \in [0, d(x, y)]$. By definition, $\gamma(0) = x$, $\gamma(d(x, y)) = y$, and $\|\gamma'(s)\|_{\gamma(s)} = 1$ for all s.

Fix an arbitrary vector $u \in \mathbb{R}^n$. We define the scalar function $Q(s) = \langle u, H(\gamma(s))u\rangle = \|u\|_{\gamma(s)}^2$, which measures the magnitude of the metric along the path. Differentiating $Q(s)$ with respect to s yields:

$$Q'(s) = \frac{d}{ds}D^2\phi(\gamma(s))[u, u] = D^3\phi(\gamma(s))[\gamma'(s), u, u]. \tag{83}$$

Applying the differential condition to bound the derivative, we obtain:

$$|Q'(s)| \leq L\|\gamma'(s)\|_{\gamma(s)}\|u\|_{\gamma(s)}^2. \tag{84}$$

Since γ is parameterized by arc length, $\|\gamma'(s)\|_{\gamma(s)} = 1$. Substituting $Q(s) = \|u\|_{\gamma(s)}^2$, we establish the differential inequality:

$$-L \cdot Q(s) \leq Q'(s) \leq L \cdot Q(s). \tag{85}$$

Dividing by $Q(s)$ and integrating from $s = 0$ to $s = d(x, y)$ gives:

$$-L \cdot d(x, y) \leq \ln(Q(d(x, y))) - \ln(Q(0)) \leq L \cdot d(x, y). \tag{86}$$

Exponentiating all terms yields:

$$e^{-L\cdot d(x,y)}\langle u, H(x)u\rangle \leq \langle u, H(y)u\rangle \leq e^{L\cdot d(x,y)}\langle u, H(x)u\rangle. \tag{87}$$

Because this inequality holds for all vectors $u \in \mathbb{R}^n$, the positive semi-definite matrix ordering directly follows:

$$e^{-L\cdot d(x,y)}H(x) \preceq H(y) \preceq e^{L\cdot d(x,y)}H(x). \tag{88}$$

Part 2: Integral Condition $\implies$ Differential Condition

Assume the integral condition holds. We evaluate the local geometry using a first-order Euclidean perturbation, which bypasses the need for geodesic parameterization. Fix $x \in \mathrm{int}(K)$ and arbitrary nonzero vectors $v, u \in \mathbb{R}^n$. For an infinitesimally small step size $\epsilon > 0$, define the point $y = x + \epsilon\frac{v}{\|v\|_x}$.

Because the first-order approximation of the Riemannian distance along any smooth path matches the local metric norm, the distance to y is $d(x, y) = \epsilon + o(\epsilon)$. Applying the upper bound of the integral condition for the chosen vector u:

$$D^2\phi\left(x + \epsilon\frac{v}{\|v\|_x}\right)[u, u] \leq e^{L\epsilon+o(\epsilon)} D^2\phi(x)[u, u]. \tag{89}$$

Rearranging to isolate the change in the Hessian quadratic form gives:

$$D^2\phi\left(x + \epsilon\frac{v}{\|v\|_x}\right)[u, u] - D^2\phi(x)[u, u] \leq \left(e^{L\epsilon+o(\epsilon)} - 1\right)\|u\|_x^2. \tag{90}$$

Dividing both sides by ϵ and taking the limit as $\epsilon \to 0$, the left-hand side converges to the directional derivative of the Hessian, while the right-hand side converges via standard Taylor expansion:

$$D^3\phi(x)\left[\frac{v}{\|v\|_x}, u, u\right] \leq L\|u\|_x^2. \tag{91}$$

Because the third derivative tensor is linear in its first argument, we multiply by the scalar $\|v\|_x$ to obtain:

$$D^3\phi(x)[v, u, u] \leq L\|v\|_x\|u\|_x^2. \tag{92}$$

By applying the exact same limit procedure to the lower bound $e^{-L\cdot d(x,y)}$, we derive the symmetric negative bound $D^3\phi(x)[v, u, u] \geq -L\|v\|_x\|u\|_x^2$. Combining these yields the strict absolute value bound $|D^3\phi(x)[v, u, u]| \leq L\|v\|_x\|u\|_x^2$, completing the proof. □

Lemma 1 (Kinematic Properties of Hessian Geodesics)**.** *Let $z(t)$ be a geodesic path parameterized by t on a Hessian manifold defined by a strictly convex barrier $\phi(z)$, and let $H(z) = \nabla^2\phi(z)$ be the local metric tensor. Let $E(t) = \|\dot{z}(t)\|_{z(t)}^2$ denote the kinetic energy of the path.*

1. ***Conservation of Kinetic Energy:*** *For any such geodesic, the kinetic energy is strictly conserved along the path ($\dot{E}(t) = 0$).*

2. ***Kinematic Acceleration Bound:*** *If ϕ satisfies L-generalized self-concordance (an equivalent consequence of L-Hessian stability), the local magnitude of the geodesic acceleration is strictly bounded by the kinetic energy:*

$$\|\ddot{z}\|_z \leq \frac{L}{2}E. \tag{93}$$

(Note: For standard self-concordant barriers where $L = 2$, this recovers exactly $\|\ddot{z}\|_z \leq E$).

Proof. **Part 1: Conservation of Kinetic Energy**
We compute the time derivative of the kinetic energy $E(t) = \langle\dot{z}, H(z)\dot{z}\rangle$ applying the standard product and chain rules:

$$\frac{d}{dt}E(t) = 2\langle\ddot{z}, H(z)\dot{z}\rangle + \langle\dot{z}, \dot{H}(z)\dot{z}\rangle. \tag{94}$$

By the definition of a geodesic on a Hessian manifold, the path satisfies the differential equation defined by the Levi-Civita connection:

$$\ddot{z} = -\frac{1}{2}H(z)^{-1}D^3\phi(z)[\dot{z}, \dot{z}, \cdot]. \tag{95}$$

We substitute Equation 95 into the first term of the derivative:

$$2\langle\ddot{z}, H(z)\dot{z}\rangle = 2\left\langle -\frac{1}{2}H(z)^{-1}D^3\phi(z)[\dot{z}, \dot{z}, \cdot], H(z)\dot{z}\right\rangle = -D^3\phi(z)[\dot{z}, \dot{z}, \dot{z}]. \tag{96}$$

For the second term, the time derivative of the Hessian matrix acting on the velocity vector is exactly the third directional derivative:

$$\langle \dot{z}, \dot{H}(z)\dot{z} \rangle = D^3\phi(z)[\dot{z}, \dot{z}, \dot{z}]. \tag{97}$$

Adding these terms together results in perfect cancellation:

$$\frac{d}{dt}E(t) = -D^3\phi(z)[\dot{z}, \dot{z}, \dot{z}] + D^3\phi(z)[\dot{z}, \dot{z}, \dot{z}] = 0. \tag{98}$$

Thus, the kinetic energy $E(t)$ remains constant for the entire duration of the geodesic step.

Part 2: Kinematic Acceleration Bound
To bound the local magnitude of the geodesic acceleration, we express its local norm using the dual supremum over the unit ball in the local metric:

$$\|\ddot{z}\|_z = \sup_{\|w\|_z \le 1} \langle \ddot{z}, H(z)w \rangle. \tag{99}$$

Substituting the geodesic equation (Equation 95) into the inner product yields:

$$\langle \ddot{z}, H(z)w \rangle = \left\langle -\frac{1}{2}H(z)^{-1}D^3\phi(z)[\dot{z}, \dot{z}, \cdot], H(z)w \right\rangle = -\frac{1}{2}D^3\phi(z)[\dot{z}, \dot{z}, w]. \tag{100}$$

Under the assumption of L-generalized self-concordance (which is equivalent to L-Hessian stability), the third derivative tensor is bounded by the local norms of its arguments:

$$|D^3\phi(z)[\dot{z}, \dot{z}, w]| \le L\|\dot{z}\|_z^2\|w\|_z. \tag{101}$$

Applying this inequality to our inner product:

$$|\langle \ddot{z}, H(z)w \rangle| \le \frac{L}{2}\|\dot{z}\|_z^2\|w\|_z. \tag{102}$$

Taking the supremum over all vectors w such that $\|w\|_z \le 1$, the right-hand side is maximized exactly at $\frac{L}{2}\|\dot{z}\|_z^2$. Since $E = \|\dot{z}\|_z^2$, we conclude:

$$\|\ddot{z}\|_z \le \frac{L}{2}E. \tag{103}$$

□

Lemma 2 (Sharp Geodesic Bound on the Jeffreys Divergence)**.** *Let z_* and z be points on a Hessian manifold with metric H satisfying global L-Hessian stability. The generalized Jeffreys divergence $V(z) = \langle z - z_*, \nabla\phi(z) - \nabla\phi(z_*) \rangle$ is strictly bounded by the Riemannian geodesic distance $d_R(z, z_*)$ according to:*

$$V(z) \le \frac{8}{L^2}\left(e^{\frac{L}{2}d_R(z,z_*)} - 1\right) - \frac{4}{L}d_R(z, z_*). \tag{104}$$

(Note: For standard self-concordance where $L = 2$, this sharply limits the divergence to $V \le 2(e^{d_R} - d_R - 1)$).

Proof. Let $\gamma(s)$ be the unit-speed true Riemannian geodesic connecting $\gamma(0) = z_*$ to $\gamma(d_R) = z$, where d_R is the geodesic distance. Since the path is parameterized by arc length, the local velocity satisfies $\|\gamma'(s)\|_{\gamma(s)} = 1$ for all $s \in [0, d_R]$.

We represent the primal and dual displacement vectors exactly as their line integrals along this geodesic:

$$z - z_* = \int_0^{d_R} \gamma'(s) ds, \tag{105}$$

$$\nabla\phi(z) - \nabla\phi(z_*) = \int_0^{d_R} H(\gamma(u))\gamma'(u) du. \tag{106}$$

We express the Jeffreys divergence by taking the inner product of these two integrals. By linearity of the inner product, this forms a double integral over the square $[0, d_R] \times [0, d_R]$:

$$V(z) = \int_0^{d_R} \int_0^{d_R} \langle \gamma'(s), H(\gamma(u))\gamma'(u)\rangle \, ds \, du. \tag{107}$$

To bound the integrand, we apply the Cauchy-Schwarz inequality pointwise in the local metric space at $\gamma(u)$:

$$\langle \gamma'(s), H(\gamma(u))\gamma'(u)\rangle \leq \|\gamma'(s)\|_{\gamma(u)} \|\gamma'(u)\|_{\gamma(u)}. \tag{108}$$

Since γ is a unit-speed geodesic, the second term is identically 1. For the first term, we invoke the global L-Hessian stability property. The metric stretch between any two points s and u on the geodesic is strictly bounded by the arc length between them, $|u - s|$:

$$H(\gamma(u)) \preceq e^{L|u-s|} H(\gamma(s)). \tag{109}$$

Therefore, the local norm of the velocity vector $\gamma'(s)$ evaluated at the point $\gamma(u)$ is bounded by:

$$\|\gamma'(s)\|^2_{\gamma(u)} = \langle \gamma'(s), H(\gamma(u))\gamma'(s)\rangle \leq e^{L|u-s|} \|\gamma'(s)\|^2_{\gamma(s)} = e^{L|u-s|}. \tag{110}$$

Taking the square root yields $\|\gamma'(s)\|_{\gamma(u)} \leq e^{\frac{L}{2}|u-s|}$. Substituting this back into Equation 107, we obtain a scalar double integral:

$$V(z) \leq \int_0^{d_R} \int_0^{d_R} e^{\frac{L}{2}|u-s|} \, ds \, du. \tag{111}$$

By symmetry, we evaluate this integral by doubling the integral over the lower triangle (where $u \geq s$ and $|u - s| = u - s$):

$$V(z) \leq 2 \int_0^{d_R} \left(\int_0^u e^{\frac{L}{2}(u-s)} \, ds \right) du. \tag{112}$$

Evaluating the inner integral with respect to s:

$$\int_0^u e^{\frac{L}{2}(u-s)} \, ds = \left[-\frac{2}{L} e^{\frac{L}{2}(u-s)} \right]_{s=0}^{s=u} = \frac{2}{L} \left(e^{\frac{L}{2}u} - 1 \right). \tag{113}$$

Substituting this back into the outer integral:

$$V(z) \leq \frac{4}{L} \int_0^{d_R} \left(e^{\frac{L}{2}u} - 1 \right) du = \frac{4}{L} \left[\frac{2}{L} e^{\frac{L}{2}u} - u \right]_0^{d_R}. \tag{114}$$

Evaluating at the boundaries yields the final, sharp bound:

$$V(z) \leq \frac{8}{L^2} \left(e^{\frac{L}{2}d_R(z,z_*)} - 1 \right) - \frac{4}{L} d_R(z, z_*). \tag{115}$$

□

Adapted Proof for Symmetric Cones. Let the domain be a symmetric cone with rank r. By the spectral theorem of Euclidean Jordan Algebras, the Riemannian geodesic γ connecting z_* to z can be perfectly diagonalized. There exists a parallel-transported, orthogonal frame of eigenvectors $\{e_1, \ldots, e_r\}$ such that the local velocity vector is constant in this frame:

$$\gamma'(t) = \sum_{i=1}^{r} \frac{\lambda_i}{d_R} e_i, \tag{116}$$

where λ_i are the eigenvalues of the displacement, and the total Riemannian distance is the Euclidean norm of the spectrum: $d_R = \sqrt{\sum_{i=1}^{r} \lambda_i^2}$.

Because the metric commutes with the tangent vectors along the geodesic, the action of the Hessian $H(\gamma(u))$ on the velocity vector $\gamma'(s)$ decomposes exactly into independent 1-dimensional exponential stretches and compressions:

$$H(\gamma(u))e_i = e^{\lambda_i \frac{u-s}{d_R}} H(\gamma(s))e_i. \tag{117}$$

We substitute this exact spectral representation into the double integral of the Jeffreys divergence:

$$V(z) = \int_0^{d_R} \int_0^{d_R} \langle \gamma'(s), H(\gamma(u))\gamma'(u) \rangle \, ds \, du. \tag{118}$$

Because the eigen-frame is strictly orthogonal in the local metric space ($\langle e_i, He_j \rangle = 0$ for $i \neq j$), all cross-terms vanish. The inner product collapses into an exact sum of scalar components:

$$\langle \gamma'(s), H(\gamma(u))\gamma'(u) \rangle = \sum_{i=1}^{r} \left(\frac{\lambda_i}{d_R} \right)^2 e^{\lambda_i \frac{u-s}{d_R}}. \tag{119}$$

We inject this exact equality back into the double integral. By Fubini's theorem, we can swap the sum and the integrals, yielding r independent 1-dimensional integrals:

$$V(z) = \sum_{i=1}^{r} \left(\frac{\lambda_i}{d_R} \right)^2 \int_0^{d_R} \int_0^{d_R} e^{\lambda_i \frac{u-s}{d_R}} \, ds \, du. \tag{120}$$

Evaluating the inner integral with respect to s:

$$\int_0^{d_R} e^{\lambda_i \frac{u-s}{d_R}} \, ds = \left[-\frac{d_R}{\lambda_i} e^{\lambda_i \frac{u-s}{d_R}} \right]_{s=0}^{s=d_R} = \frac{d_R}{\lambda_i} \left(e^{\lambda_i \frac{u}{d_R}} - e^{\lambda_i \frac{u-d_R}{d_R}} \right). \tag{121}$$

Evaluating the outer integral with respect to u:

$$\frac{\lambda_i}{d_R} \int_0^{d_R} \left(e^{\lambda_i \frac{u}{d_R}} - e^{\lambda_i \frac{u-d_R}{d_R}} \right) du = \left[e^{\lambda_i \frac{u}{d_R}} - e^{\lambda_i \frac{u-d_R}{d_R}} \right]_0^{d_R} = e^{\lambda_i} - 1 - (1 - e^{-\lambda_i}). \tag{122}$$

This beautifully simplifies to $e^{\lambda_i} + e^{-\lambda_i} - 2$, which is exactly $2(\cosh(\lambda_i) - 1)$. Thus, the divergence evaluates to the exact equality:

$$V(z) = \sum_{i=1}^{r} 2(\cosh(\lambda_i) - 1). \tag{123}$$

Finally, to bound this sum by the total Riemannian distance d_R, we note that the function $f(x) = 2(\cosh(x) - 1)$ is strictly convex, monotonically increasing for $x \geq 0$, and grows faster than quadratically. Therefore, for any vector $\lambda \in \mathbb{R}^r$, the sum of the functions of its components is strictly bounded by the function of its Euclidean norm:

$$\sum_{i=1}^{r} 2(\cosh(\lambda_i) - 1) \leq 2 \left(\cosh \left(\sqrt{\sum_{i=1}^{r} \lambda_i^2} \right) - 1 \right). \tag{124}$$

Since $d_R = \sqrt{\sum_{i=1}^{r} \lambda_i^2}$, we recover the exact, sharp symmetric bound:

$$V(z) \leq 2(\cosh(d_R(z, z_*)) - 1). \tag{125}$$

□

Lemma 3 (Curvature Bounds of the Jeffreys Divergence)**.** *Let V be the Jeffreys divergence, $E = \|\dot{z}\|_z^2$ be the local kinetic energy, and U be the centering residual. Under the standard self-concordant inequalities, the second derivative of the divergence along the geodesic, $\ddot{V} \leq 2E + E\|U\|_z$, admits three strict upper bounds depending on the geometric regime:*

1. ***Global Relaxed Bound (Far-Field):*** *By applying the triangle inequality to U and using subadditivity ($\sqrt{V^2 + 4V} \leq V + 2\sqrt{V}$), we obtain:*

$$\ddot{V}_{relax} \leq 2E(1 + V + \sqrt{V}). \tag{126}$$

2. ***Global Exact Bound (Mid-Field):*** *Using the exact root of the self-concordant inequality $V \geq r^2/(1 + r)$ without subadditive relaxation yields a tighter global bound:*

$$\ddot{V}_{exact} \leq E\left(2 + V + \sqrt{V^2 + 4V}\right). \tag{127}$$

3. ***Local Unrelaxed Bound (Near-Field):*** *For $E < 1/4$, the target is deep inside the Dikin ellipsoid. Bypassing the triangle inequality entirely, standard self-concordant theory bounds the residual exactly by $\|U\|_z \leq \frac{r^2}{1-r}$, where $r \leq \frac{\sqrt{E}}{1-\sqrt{E}}$. This yields the purely local metric bound:*

$$\ddot{V}_{local} \leq E\left(2 + \frac{E}{(1 - \sqrt{E})(1 - 2\sqrt{E})}\right). \tag{128}$$

Theorem 12 (Kinematic Bound on Divergence Curvature)**.** *Let $z(t)$ be a geodesic parameterized by arc length on the Hessian manifold defined by a self-concordant barrier ϕ, with constant kinetic energy $E = \|\dot{z}\|_z^2$. Let z_* be an arbitrary target point in the domain. Assuming the kinematic stability axiom $\|\ddot{z}\|_z \leq E$, the second time derivative of the generalized Jeffreys divergence $V(t) = \langle z - z_*, \nabla\phi(z) - \nabla\phi(z_*)\rangle$ is bounded by:*

$$\ddot{V} \leq 2E + E\|U\|_z \tag{129}$$

where $U = (z - z_*) - H(z)^{-1}(\nabla\phi(z) - \nabla\phi(z_*))$ is the aggregated residual vector.

Proof. We begin by taking the first time derivative of the generalized Jeffreys divergence $V(z) = \langle z - z_*, \nabla\phi(z) - \nabla\phi(z_*)\rangle$ along the geodesic $z(t)$. Since z_* and $\nabla\phi(z_*)$ are fixed constants, their time derivatives are zero:

$$\dot{V} = \langle \dot{z}, \nabla\phi(z) - \nabla\phi(z_*)\rangle + \langle z - z_*, H(z)\dot{z}\rangle. \tag{130}$$

Taking the second time derivative yields:

$$\begin{aligned}\ddot{V} = {} & \langle \ddot{z}, \nabla\phi(z) - \nabla\phi(z_*)\rangle + \langle \dot{z}, H(z)\dot{z}\rangle \\ & + \langle \dot{z}, H(z)\dot{z}\rangle + \langle z - z_*, \dot{H}(z)\dot{z}\rangle + \langle z - z_*, H(z)\ddot{z}\rangle.\end{aligned} \tag{131}$$

Recognizing that the kinetic energy is $E = \langle \dot{z}, H(z)\dot{z}\rangle$, we substitute $2E$ and group the acceleration terms:

$$\ddot{V} = 2E + \langle \ddot{z}, \nabla\phi(z) - \nabla\phi(z_*) + H(z)(z - z_*)\rangle + D^3\phi(z)[\dot{z}, \dot{z}, z - z_*]. \tag{132}$$

By the definition of a geodesic on a Hessian manifold, the acceleration $\ddot{z}$ is driven by the Levi-Civita connection:

$$\ddot{z} = -\frac{1}{2}H(z)^{-1}D^3\phi(z)[\dot{z}, \dot{z}, \cdot]. \tag{133}$$

Applying this identity to the vector $w = z - z_*$ allows us to rewrite the third derivative term as a local inner product with the acceleration:

$$D^3\phi(z)[\dot{z}, \dot{z}, z - z_*] = -2\langle \ddot{z}, H(z)(z - z_*)\rangle. \tag{134}$$

Substituting this back into Equation 132 results in the cancellation of one of the primal residual terms:

$$\begin{aligned}\ddot{V} &= 2E + \langle \ddot{z}, \nabla\phi(z) - \nabla\phi(z_*) + H(z)(z - z_*)\rangle - 2\langle \ddot{z}, H(z)(z - z_*)\rangle \\ &= 2E + \langle \ddot{z}, (\nabla\phi(z) - \nabla\phi(z_*)) - H(z)(z - z_*)\rangle.\end{aligned} \tag{135}$$

We factor out $-H(z)$ from the right side of the inner product to reveal the generalized aggregated residual U:

$$\begin{aligned}\ddot{V} &= 2E - \langle \ddot{z}, H(z)\left[(z - z_*) - H(z)^{-1}(\nabla\phi(z) - \nabla\phi(z_*))\right]\rangle \\ &= 2E - \langle \ddot{z}, U\rangle_z.\end{aligned} \tag{136}$$

To strictly bound this geometric drift, we apply the Cauchy-Schwarz inequality in the local metric space:

$$|\langle \ddot{z}, U\rangle_z| \leq \|\ddot{z}\|_z\|U\|_z. \tag{137}$$

Finally, invoking the kinematic stability axiom $\|\ddot{z}\|_z \leq E$, we obtain the upper bound:

$$\ddot{V} \leq 2E + E\|U\|_z. \tag{138}$$

□

Lemma 4 (Global Bound on the Residual and Divergence Drift)**.** *Let ϕ be a standard self-concordant barrier (where $L = 2$). Let z_* be an arbitrary target point, and let $V(z) = \langle z - z_*, \nabla\phi(z) - \nabla\phi(z_*)\rangle$ be the Jeffreys divergence. Consider the aggregated centering residual $U = (z - z_*) - H(z)^{-1}(\nabla\phi(z) - \nabla\phi(z_*))$.*

The local norm of the residual is strictly bounded by:

$$\|U\|_z \leq 2V(z) + 2\sqrt{V(z)}. \tag{139}$$

Consequently, along a Riemannian geodesic $z(t)$ with constant kinetic energy $E = \|\dot{z}\|_z^2$ satisfying the kinematic stability axiom $\|\ddot{z}\|_z \leq E$, the second time derivative of the divergence is globally bounded by:

$$\ddot{V} \leq 2E\left(1 + \sqrt{V(z)} + V(z)\right). \tag{140}$$

Proof. We begin by bounding the local magnitude of the aggregated residual vector U. Applying the triangle inequality in the local metric space at z:

$$\|U\|_z = \|(z-z_*)-H(z)^{-1}(\nabla\phi(z)-\nabla\phi(z_*))\|_z \leq \|z-z_*\|_z+\|H(z)^{-1}(\nabla\phi(z)-\nabla\phi(z_*))\|_z. \tag{141}$$

Recognizing that the second term is exactly the local dual norm of the gradient difference, we have:

$$\|U\|_z \leq \|z - z_*\|_z + \|\nabla\phi(z) - \nabla\phi(z_*)\|_z^*. \tag{142}$$

To bound these local norms using the Jeffreys divergence V, we rely on the fundamental primal-dual inner product inequalities for standard self-concordant functions. Let $r = \|z - z_*\|_z$ be the local primal distance. The divergence strictly bounds r according to:

$$V \geq \frac{r^2}{1 + r}. \tag{143}$$

Rearranging this into a standard quadratic inequality yields $r^2 - Vr - V \leq 0$. By the quadratic formula, the positive root is bounded by:

$$r \leq \frac{V + \sqrt{V^2 + 4V}}{2}. \tag{144}$$

By the subadditivity of the square root function for non-negative operands, $\sqrt{V^2 + 4V} \leq \sqrt{V^2} + \sqrt{4V} = V + 2\sqrt{V}$. Substituting this upper bound establishes a strict limit on the primal distance that correctly vanishes at $V = 0$:

$$\|z - z_*\|_z \leq \frac{V + (V + 2\sqrt{V})}{2} = V + \sqrt{V}. \tag{145}$$

By the geometric symmetry of the self-concordant Legendre transform, the exact same bounding logic applies to the local dual distance $g = \|\nabla\phi(z) - \nabla\phi(z_*)\|_z^*$. The divergence satisfies $V \geq \frac{g^2}{1+g}$, which via identical algebraic simplification is bounded as:

$$\|\nabla\phi(z) - \nabla\phi(z_*)\|_z^* \leq V + \sqrt{V}. \tag{146}$$

Substituting these two bounds back into the triangle inequality (Equation 142) yields the strict global bound on the residual:

$$\|U\|_z \leq (V + \sqrt{V}) + (V + \sqrt{V}) = 2V + 2\sqrt{V}. \tag{147}$$

Finally, we project this residual bound onto the continuous geodesic path. The second time derivative of the divergence along the geodesic is given by:

$$\ddot{V} = 2E - \langle \ddot{z}, U\rangle_z. \tag{148}$$

Applying the Cauchy-Schwarz inequality to the local inner product:

$$\ddot{V} \leq 2E + \|\ddot{z}\|_z \|U\|_z. \tag{149}$$

Invoking the kinematic stability axiom for standard self-concordant barriers ($\|\ddot{z}\|_z \leq E$) and substituting our bound from Equation 147:

$$\ddot{V} \leq 2E + E(2V + 2\sqrt{V}) = 2E(1 + \sqrt{V} + V). \tag{150}$$

□

Lemma 5 (Exact Geodesic Acceleration of the Jeffreys Divergence)**.** *Let $V(z) = \langle z - z_*, \nabla\phi(z) - \nabla\phi(z_*)\rangle$ be the Jeffreys divergence, $E = \|\dot{z}\|_z^2$ be the local kinetic energy, and $U = (z - z_*) - H(z)^{-1}(\nabla\phi(z) - \nabla\phi(z_*))$ be the centering residual. For a trajectory $z(t)$ following the natural geodesic equation on the Hessian manifold induced by ϕ, the second time derivative of the divergence is exactly:*

$$\ddot{V} = 2E - \langle \ddot{z}, U\rangle_z. \tag{151}$$

Proof. Let $v = z - z_*$ and $g = \nabla\phi(z) - \nabla\phi(z_*)$, such that $V = \langle v, g\rangle$. Differentiating with respect to the trajectory parameter t yields the first derivative:

$$\dot{V} = \langle \dot{v}, g\rangle + \langle v, \dot{g}\rangle. \tag{152}$$

Since $\dot{v} = \dot{z}$ and $\dot{g} = \frac{d}{dt}\nabla\phi(z) = H(z)\dot{z}$, we have $\dot{V} = \langle \dot{z}, g\rangle + \langle v, H(z)\dot{z}\rangle$. Differentiating a second time yields:

$$\ddot{V} = (\langle \ddot{z}, g\rangle + \langle \dot{z}, \dot{g}\rangle) + \left(\langle \dot{v}, H(z)\dot{z}\rangle + \langle v, \dot{H}(z)\dot{z}\rangle + \langle v, H(z)\ddot{z}\rangle\right). \tag{153}$$

We evaluate the first-order inner products using $\dot{v} = \dot{z}$ and $\dot{g} = H(z)\dot{z}$:

$$\langle \dot{z}, \dot{g}\rangle = \langle \dot{z}, H(z)\dot{z}\rangle = E \tag{154}$$

$$\langle \dot{v}, H(z)\dot{z}\rangle = \langle \dot{z}, H(z)\dot{z}\rangle = E. \tag{155}$$

These terms sum exactly to $2E$. For the tensor derivative term, we use the definition of the Hessian directional derivative along the path: $\langle v, \dot{H}(z)\dot{z}\rangle = D^3\phi(z)[\dot{z}, \dot{z}, v]$. Because $z(t)$ is a geodesic on the Hessian manifold, it satisfies the length-minimizing condition $\ddot{z} + \frac{1}{2}H(z)^{-1}D^3\phi(z)[\dot{z}, \dot{z}] = 0$, yielding the identity:

$$D^3\phi(z)[\dot{z}, \dot{z}] = -2H(z)\ddot{z}. \tag{156}$$

Substituting this directly into the inner product gives:

$$\langle v, \dot{H}(z)\dot{z}\rangle = \langle v, -2H(z)\ddot{z}\rangle = -2\langle v, H(z)\ddot{z}\rangle. \tag{157}$$

Plugging these simplified terms back into Equation 153 yields:

$$\ddot{V} = 2E + \langle \ddot{z}, g\rangle + \langle v, H(z)\ddot{z}\rangle - 2\langle v, H(z)\ddot{z}\rangle = 2E + \langle \ddot{z}, g\rangle - \langle v, H(z)\ddot{z}\rangle. \tag{158}$$

By the symmetry of the Hessian matrix, $\langle v, H(z)\ddot{z}\rangle = \langle H(z)v, \ddot{z}\rangle$. Factoring out the $\ddot{z}$ vector from the remaining inner products gives:

$$\ddot{V} = 2E + \langle \ddot{z}, g - H(z)v\rangle. \tag{159}$$

To convert this to the local metric geometry $\langle a, b\rangle_z = \langle a, H(z)b\rangle$, we factor $H(z)$ out of the right-hand side:

$$g - H(z)v = H(z)(H(z)^{-1}g - v). \tag{160}$$

By the definition of the centering residual, $U = v - H(z)^{-1}g$, meaning $(H(z)^{-1}g - v) = -U$. Substituting this back into the standard Euclidean inner product produces the local metric formulation, completing the proof:

$$\ddot{V} = 2E + \langle \ddot{z}, H(z)(-U)\rangle = 2E - \langle \ddot{z}, U\rangle_z. \tag{161}$$

□

Definition 5 (Newton Target Tangent)**.** *Let z be the current state on the manifold and z_* be the target state. Let $g = \nabla\phi(z) - \nabla\phi(z_*)$ denote the dual residual vector. The geodesic velocity (tangent) $\dot{z}$ is defined as the pure local Newton direction minimizing the dual residual:*

$$\dot{z} = -H(z)^{-1}(\nabla\phi(z) - \nabla\phi(z_*)) = -H(z)^{-1}g. \tag{162}$$

By definition, this tangent induces a local kinetic energy $E = \|\dot{z}\|_z^2 = \langle \dot{z}, H(z)\dot{z}\rangle$, which is mathematically equivalent to the squared Newton decrement $E = \|g\|_z^{*2}$.

Lemma 6 (Exact Derivative of the Divergence)**.** *Let $V(z) = \langle z - z_*, \nabla\phi(z) - \nabla\phi(z_*)\rangle$ be the Jeffreys divergence between the current state and the target. Under the Newton Target Tangent $\dot{z}$, the continuous time derivative of the divergence is perfectly locked to the sum of the current divergence and the local kinetic energy:*

$$\dot{V} = -(E + V). \tag{163}$$

Proof. We differentiate the Jeffreys divergence $V = \langle z - z_*, \nabla\phi(z) - \nabla\phi(z_*)\rangle$ with respect to the continuous path parameter t. Applying the standard product rule for inner products yields:

$$\dot{V} = \langle \dot{z}, \nabla\phi(z) - \nabla\phi(z_*)\rangle + \left\langle z - z_*, \frac{d}{dt}\nabla\phi(z)\right\rangle. \tag{164}$$

By the chain rule, the time derivative of the gradient along the trajectory is exactly the Hessian-vector product: $\frac{d}{dt}\nabla\phi(z) = H(z)\dot{z}$. Substituting this into Equation 164 and applying the symmetry of the Hessian $H(z)$ to the second term gives:

$$\dot{V} = \langle \dot{z}, g\rangle + \langle z - z_*, H(z)\dot{z}\rangle. \tag{165}$$

From Definition 5, the chosen tangent satisfies $H(z)\dot{z} = -g$. We substitute this identity into both terms of Equation 165:

1. For the first term: $\langle \dot{z}, g\rangle = \langle \dot{z}, -H(z)\dot{z}\rangle = -\|\dot{z}\|_z^2 = -E$.
2. For the second term: $\langle z - z_*, H(z)\dot{z}\rangle = \langle z - z_*, -g\rangle = -V$.

Combining these results mathematically proves that the exact rate of descent of the Jeffreys divergence is dictated purely by the native geometric quantities:

$$\dot{V} = -E - V = -(E + V). \tag{166}$$

□

Lemma 7 (Curvature Bounds of the Jeffreys Divergence)**.** *Let V be the Jeffreys divergence, $E = \|\dot{z}\|_z^2$ be the local kinetic energy, and U be the centering residual. Under the standard self-concordant inequalities, the second derivative of the divergence along the geodesic, $\ddot{V} \leq 2E + E\|U\|_z$, admits three strict upper bounds depending on the geometric regime:*

1. ***Global Relaxed Bound (Far-Field):*** *By applying the triangle inequality to U and using subadditivity ($\sqrt{V^2 + 4V} \leq V + 2\sqrt{V}$), we obtain:*

$$\ddot{V}_{relax} \leq 2E(1 + V + \sqrt{V}). \tag{167}$$

2. ***Global Exact Bound (Mid-Field):*** *Using the exact root of the self-concordant inequality $V \geq r^2/(1 + r)$ without subadditive relaxation yields a tighter global bound:*

$$\ddot{V}_{exact} \leq E\left(2 + V + \sqrt{V^2 + 4V}\right). \tag{168}$$

3. ***Local Unrelaxed Bound (Near-Field):*** *For $E < 1/4$, the target is deep inside the Dikin ellipsoid. Bypassing the triangle inequality entirely, standard self-concordant theory bounds the residual exactly by $\|U\|_z \leq \frac{r^2}{1-r}$, where $r \leq \frac{\sqrt{E}}{1-\sqrt{E}}$. This yields the purely local metric bound:*

$$\ddot{V}_{local} \leq E\left(2 + \frac{E}{(1 - \sqrt{E})(1 - 2\sqrt{E})}\right). \tag{169}$$

Theorem 13 (Global Convergence and Rates of the Discrete Geodesic Step)**.** *Consider a geodesic trajectory parameterized by kinetic energy E, with velocity satisfying $\dot{V} = -(E+V)$, and kinematic stability $\|\ddot{z}\|_z \le E$. We define the discrete step-size policy:*

$$h(E) = \begin{cases} \frac{1}{2E} & \textit{if } E \ge 1 \quad \textit{(Far-Field)} \\ \frac{1}{2\sqrt{E}} & \textit{if } 1/4 \le E < 1 \quad \textit{(Mid-Field)} \\ 1 & \textit{if } E < 1/4 \quad \textit{(Near-Field)} \end{cases} \tag{170}$$

This policy guarantees strict monotonic descent ($\Delta V < 0$) across all regimes, achieving a linear contraction rate in the Far-Field and a pure quadratic convergence rate ($V(h) \le \mathcal{O}(V(0)^2)$) in the Near-Field.

Proof. We evaluate the Taylor surrogate $V(h) \le V - h(E+V) + \frac{1}{2}h^2\ddot{V}_{upper}$ across the three regimes.

Case 1: The Far-Field ($E \ge 1$).
Using the Global Relaxed Bound ($\ddot{V}_{relax}$) and $h = \frac{1}{2E}$, the surrogate simplifies to:

$$\Delta V \le -\frac{E+V}{2E} + \frac{1}{4E^2}E(V+\sqrt{V}+1) = \frac{1}{4E}\left(1 - 2E + \sqrt{V} - V\right). \tag{171}$$

The expression $(\sqrt{V} - V)$ attains its global maximum of 0.25 at $V = 1/4$. Thus, the bound is universally limited by $\Delta V \le \frac{1}{4E}(1.25 - 2E)$. Because $E \ge 1$, we have $(1.25 - 2E) \le -0.75$, yielding $\Delta V \le -\frac{0.75}{4E} < 0$. Strict linear descent is guaranteed.

Case 2: The Mid-Field ($1/4 \le E < 1$).
We apply the Global Exact Bound ($\ddot{V}_{exact}$) to prevent over-estimation in the transition zone, evaluating the surrogate at a constant $h = 1/2$:

Substituting $h = 1/(2\sqrt{E})$ into the exact Taylor surrogate, strict descent requires:

$$\sqrt{E}(2 + V + \sqrt{V^2 + 4V}) < 4E + 4V. \tag{172}$$

Let $x = \sqrt{E}$. We rearrange the inequality to isolate the square root term:

$$x\sqrt{V^2 + 4V} < 4x^2 - 2x + V(4 - x). \tag{173}$$

Because the Mid-Field restricts $x \ge 0.5$, the right-hand side is strictly positive ($4x^2 - 2x \ge 0$ and $4 - x > 0$). Therefore, we can safely square both sides:

$$x^2(V^2 + 4V) < (4x^2 - 2x)^2 + 2V(4-x)(4x^2 - 2x) + V^2(4-x)^2. \tag{174}$$

Subtracting $x^2(V^2+4V)$ from both sides and grouping by powers of V yields a simple quadratic inequality:

$$8(2-x)V^2 + 8x(4x - x^2 - 2)V + 4x^2(2x-1)^2 > 0. \tag{175}$$

This is an upward-facing parabola in V. By standard self-concordant geometry, the divergence is strictly bounded below by $V \ge \frac{x^2}{1+x}$.

Substituting this minimum bound into the quadratic yields the condition:

$$\frac{4x^2}{(1+x)^2}\left(2x^4 + 8x^3 + 5x^2 - 6x + 1\right) > 0. \tag{176}$$

For the entire Mid-Field interval ($x \in [0.5, 1)$), the polynomial $P(x) = 2x^4 + 8x^3 + 5x^2 - 6x + 1$ is strictly positive. (It attains its minimum value on this interval at $x = 0.5$, where $P(0.5) = 0.375$). Because the quadratic in V is positive at the manifold boundary and strictly increasing for valid states, descent is universally guaranteed.

Case 3: The Near-Field ($E < 1/4$).
Entering the deep quadratic basin, we take a full step $h = 1$ and apply the Local Unrelaxed Bound ($\ddot{V}_{local}$). The continuous linear terms exactly annihilate the base state:

$$V(1) \leq V(0) - (E + V(0)) + \frac{1}{2}\ddot{V}_{local} = -E + E + \frac{1}{2}\left(\frac{E^2}{(1-\sqrt{E})(1-2\sqrt{E})}\right). \tag{177}$$

Using the general inequality $E \leq V(1+\sqrt{E})$ and the assumption $E < 1/4$, we conclude that in the near-field,

$$E \leq 1.5V(0) \tag{178}$$

Substituting into the previous inequality proves $V(1)\mathcal{O}V(0)^2$.

□

Theorem 14 (Global Convergence of the Exact Rescaled Geodesic Step)**.** *Let V be the Jeffreys divergence and E be the local kinetic energy. For a geodesic parameterized by the continuous step-size policy:*

$$h(E) = \begin{cases} \frac{3}{4E} & \textit{if } E \geq 1 \quad \textit{(Far-Field)} \\ \frac{3}{2(1+\sqrt{E})} & \textit{if } 1/4 \leq E < 1 \quad \textit{(Mid-Field)} \\ 1 & \textit{if } E < 1/4 \quad \textit{(Near-Field)} \end{cases} \tag{179}$$

The algorithm exhibits C^0 continuity across all boundaries and guarantees strict monotonic descent ($\Delta V < 0$) for all $E \geq 1/4$, seamlessly transitioning to pure quadratic convergence when $E < 1/4$.

Proof. To guarantee descent, we evaluate the exact Taylor surrogate $\Delta V \leq -h(E+V) + \frac{1}{2}h^2\ddot{V}_{exact}$, requiring the curvature penalty to be strictly bounded by the linear descent:

$$\frac{1}{2}h^2E\left(2 + V + \sqrt{V^2+4V}\right) \leq h(E+V). \tag{180}$$

Part 1: The Far-Field ($E \geq 1$)
Substitute $h = \frac{3}{4E}$ into Equation 180 and divide by h:

$$\frac{3}{8E}E\left(2 + V + \sqrt{V^2+4V}\right) \leq E + V. \tag{181}$$

Multiply by 8 and isolate the square root term:

$$3\sqrt{V^2+4V} \leq 8E + 8V - 6 - 3V = 8E - 6 + 5V. \tag{182}$$

Because we are in the Far-Field ($E \geq 1$), the absolute minimum of the right side occurs at $E = 1$, yielding $3\sqrt{V^2+4V} \leq 2 + 5V$. Since $V \geq 0$, both sides are strictly positive, allowing us to square them:

$$9(V^2 + 4V) \leq 4 + 20V + 25V^2. \tag{183}$$

Subtracting the left side from the right yields:

$$16V^2 - 16V + 4 \geq 0 \implies 4(2V-1)^2 \geq 0. \tag{184}$$

This perfect square is universally non-negative for all V. Thus, strict descent is mathematically guaranteed everywhere in the Far-Field.

Part 2: The Mid-Field ($1/4 \leq E < 1$)

Substitute $h = \frac{3}{2(1+\sqrt{E})}$ into Equation 180. Letting $x = \sqrt{E} \in [0.5, 1)$, we require:

$$\frac{3x^2}{4(1+x)} \left(2 + V + \sqrt{V^2 + 4V}\right) \leq x^2 + V. \tag{185}$$

Multiply by $\frac{4(1+x)}{x^2}$ and isolate the square root:

$$3\sqrt{V^2 + 4V} \leq \frac{4(1+x)}{x^2}(x^2 + V) - 3(2 + V). \tag{186}$$

We split the fraction into $\left(\frac{4}{x^2} + \frac{4}{x}\right)$ and distribute it across $(x^2 + V)$:

$$\begin{aligned} 3\sqrt{V^2 + 4V} &\leq \left(4 + \frac{4V}{x^2} + 4x + \frac{4V}{x}\right) - 6 - 3V \\ 3\sqrt{V^2 + 4V} &\leq (4x - 2) + V\left(\frac{4}{x^2} + \frac{4}{x} - 3\right). \end{aligned} \tag{187}$$

Define the coefficients $A(x) = 4x - 2$ and $B(x) = \frac{4}{x^2} + \frac{4}{x} - 3$. For the interval $x \in [0.5, 1)$, the coefficient $A(x)$ ranges strictly from 0 to 2, and $B(x)$ ranges from 21 down to 5. Since $A(x) \geq 0$, $B(x) > 0$, and $V \geq 0$, the right-hand side is universally non-negative, allowing us to safely square both sides:

$$9V^2 + 36V \leq A^2 + 2ABV + B^2V^2. \tag{188}$$

Rearranging into a standard quadratic form in terms of V:

$$P(V) = (B^2 - 9)V^2 + (2AB - 36)V + A^2 \geq 0. \tag{189}$$

At the upper boundary $x \to 1$, we have $A = 2$ and $B = 5$. The polynomial becomes $16V^2 - 16V + 4 = 4(2V - 1)^2 \geq 0$, which is the exact perfect square from the Far-Field.

At the lower boundary $x = 0.5$, we have $A = 0$ and $B = 21$. The polynomial becomes:

$$(441 - 9)V^2 - 36V \geq 0 \implies 432V^2 - 36V \geq 0. \tag{190}$$

This factors to $36V(12V-1) \geq 0$, which is strictly satisfied for all $V \geq 1/12$. The self-concordant geometry of the manifold strictly mandates $V \geq \frac{E}{1+\sqrt{E}}$. At $E = 1/4$, this structural minimum is $V \geq 1/6$. Since $1/6 > 1/12$, the descent condition is strictly guaranteed at the boundary.

Because the leading coefficient $(B(x)^2 - 9) \geq 16 > 0$ across the entire interval, $P(V)$ is an upward-facing parabola whose required minimum root is bounded safely below the geometric reality of the manifold ($V \geq \frac{x^2}{1+x}$). Therefore, the exact continuous step size guarantees strict monotonic descent universally across the Mid-Field.

Case 3: The Near-Field ($E < 1/4$).
Entering the deep quadratic basin, we take a full step $h = 1$ and apply the Local Unrelaxed Bound ($\ddot{V}_{local}$). The continuous linear terms exactly annihilate the base state:

$$V(1) \leq V(0) - (E + V(0)) + \frac{1}{2}\ddot{V}_{local} = -E + E + \frac{1}{2}\left(\frac{E^2}{(1-\sqrt{E})(1-2\sqrt{E})}\right). \tag{191}$$

Using the general inequality $E \leq V(1 + \sqrt{E})$ and the assumption $E < 1/4$, we conclude that in the near-field,

$$E \leq 1.5V(0) \tag{192}$$

Substituting into the previous inequality proves $V(1)\mathcal{O}V(0)^2$. □

16.1 Second version

Lemma 8 (The Coupled Integral Bound for Self-Concordant Remainder). *Let F be a standard self-concordant function. For any base point z and a target point y strictly inside the local Dikin ellipsoid centered at z (such that the local Newton distance is $r = \|y - z\|_z < 1$), let $V(z, y)$ denote the Jeffreys divergence and $R(z, y)$ denote the exact Taylor remainder of the gradient map:*

$$V(z, y) = \langle y - z, \nabla F(y) - \nabla F(z)\rangle \tag{193}$$

$$R(z, y) = \nabla F(y) - \nabla F(z) - \nabla^2 F(z)(y - z) \tag{194}$$

Then, the dual local norm of the remainder is strictly bounded by the divergence:

$$\|R(z, y)\|_z^* \le 2V(z, y) \tag{195}$$

Proof. Let $h = y - z$. We parameterize the affine line segment connecting z to y as $x_s = z + sh$ for $s \in [0, 1]$.

The Jeffreys divergence can be expressed as the exact path integral of the local Hessian norm along this affine segment:

$$V(z, y) = \int_0^1 \langle h, \nabla^2 F(x_s)h\rangle\, ds = \int_0^1 \|h\|_{x_s}^2\, ds \tag{196}$$

Similarly, applying Taylor's theorem with the integral remainder to the gradient map yields the exact path integral for R:

$$R(z, y) = \int_0^1 (1 - s)\nabla^3 F(x_s)[h, h, \cdot]\, ds \tag{197}$$

We wish to bound the dual local norm at the base point, $\|R\|_z^*$. Applying Minkowski's inequality for integrals gives:

$$\|R\|_z^* \le \int_0^1 (1 - s)\|\nabla^3 F(x_s)[h, h, \cdot]\|_z^*\, ds \tag{198}$$

Because y is strictly inside the Dikin ellipsoid ($r < 1$), the affine path x_s never exits the valid domain of the base metric. By standard self-concordant metric properties, the dual norm evaluated at the base point z is bounded by the dual norm at the shifted point x_s:

$$\|\cdot\|_z^* \le \frac{1}{1 - sr}\|\cdot\|_{x_s}^* \tag{199}$$

Substituting this metric shift into Equation 198, and applying the classical local definition of standard self-concordance ($\|\nabla^3 F(x_s)[h, h, \cdot]\|_{x_s}^* \le 2\|h\|_{x_s}^2$), we obtain:

$$\|R\|_z^* \le \int_0^1 \frac{1 - s}{1 - sr}\left(2\|h\|_{x_s}^2\right) ds = 2\int_0^1 \left(\frac{1 - s}{1 - sr}\right)\|h\|_{x_s}^2\, ds \tag{200}$$

For any $r < 1$ and $s \in [0, 1]$, the rational scaling function $f(s) = \frac{1-s}{1-sr}$ is strictly decreasing. Its global maximum over the interval occurs at $s = 0$, where $f(0) = 1$. Therefore, we can bound the fraction by 1 and pull it out of the integral:

$$\|R\|_z^* \le 2(1)\int_0^1 \|h\|_{x_s}^2\, ds \tag{201}$$

Recognizing the remaining integral as the exact definition of the Jeffreys divergence completes the proof:

$$\|R(z, y)\|_z^* \le 2V(z, y) \tag{202}$$

□

Theorem 15 (Pure Quadratic Convergence Cascade)**.** *Assume the local kinetic energy of the algorithm satisfies $E < 1/4$. This kinematic condition strictly implies the target lies within the local Dikin ellipsoid ($r < 1$).*

Under this condition, taking a full geodesic Newton step ($h = 1$) guarantees immediate strict contraction of the Jeffreys divergence. The subsequent state V_{next} is governed by a dual-bound system, satisfying both an immediate linear contraction and a pure quadratic asymptote:

$$V_{next} < \frac{1}{4}V \tag{203}$$

$$V_{next} < \frac{3}{2}V^2 \leq \mathcal{O}(V^2) \tag{204}$$

Proof. By the exact definition of the Jeffreys divergence $V = \langle z - z_*, \nabla F(z) - \nabla F(z_*)\rangle$, the first derivative of the divergence along the Newton-directed geodesic ($\dot{z} = -H^{-1}g$) decomposes perfectly into the kinetic energy and the linear distance:

$$\dot{V} = \langle \dot{z}, g\rangle + \langle z - z_*, H\dot{z}\rangle = -E - V. \tag{205}$$

For the second derivative, we start with the fundamental geodesic identity:

$$\ddot{V} \leq 2E + \langle \ddot{z}, R\rangle. \tag{206}$$

Applying the local Cauchy-Schwarz inequality and the standard geodesic acceleration bound ($\|\ddot{z}\|_z \leq E$):

$$\ddot{V} \leq 2E + E\|R\|_z^*. \tag{207}$$

Because the hypothesis $E < 1/4$ mathematically guarantees the Newton distance satisfies $r < \frac{\sqrt{E}}{1-\sqrt{E}} < 1$, the state is strictly inside the local Dikin ellipsoid. Thus, we may apply the Coupled Integral Lemma ($\|R\|_z^* \leq 2V$) to completely eliminate the remainder norm:

$$\ddot{V} \leq 2E + E(2V) = 2E(1 + V). \tag{208}$$

We now evaluate the exact continuous Taylor surrogate for a full geodesic step ($h = 1$):

$$V_{next} \leq V + \dot{V}(1) + \frac{1}{2}\ddot{V}(1)^2 \tag{209}$$

$$\leq V + (-E - V) + \frac{1}{2}\left[2E(1 + V)\right]. \tag{210}$$

Distributing the second-order curvature perfectly cancels both the linear descent baseline ($-E$) and the current divergence (V), yielding the core transitional map:

$$V_{next} \leq -E + E(1 + V) = -E + E + EV = E \cdot V. \tag{211}$$

Phase 1: The Strict Linear Contraction

Applying the hypothesis $E < 1/4$ directly to Equation 211 guarantees an immediate, aggressive geometric decrease, regardless of the magnitude of V:

$$V_{next} < \frac{1}{4}V. \tag{212}$$

Phase 2: The Pure Quadratic Cascade

To decouple the metrics and formulate a pure contraction map in terms of V, we rely on the foundational geometric inequality $V \geq \frac{E}{1+\sqrt{E}}$. Rearranging this yields a strict upper bound on the local kinetic energy: $E \leq V(1 + \sqrt{E})$.

Substituting this clean bound into Equation 211 yields:

$$V_{next} \leq \left[V(1+\sqrt{E})\right] V = V^2(1+\sqrt{E}). \tag{213}$$

Applying the hypothesis $E < 1/4$, which dictates that the velocity is strictly bounded by $\sqrt{E} < 1/2$, completes the pure quadratic bound:

$$V_{next} < V^2\left(1+\frac{1}{2}\right) = \frac{3}{2}V^2. \tag{214}$$

Conclusion

Upon entering the domain $E < 1/4$, the linear bound ($V_{next} < V/4$) unconditionally drives the divergence downward. Once the divergence crosses the threshold where $\frac{3}{2}V^2 < \frac{1}{4}V$ (specifically $V < 1/6$), the quadratic bound dominates, and the algorithm permanently locks into a pure $\mathcal{O}(V^2)$ convergence cascade. $\square$